

GEN-3 SAFE HELICOPTER

DOCUMENT 00

14-TON / 8-ROTOR SAFETY-CENTERED BASELINE

Central Protected Fixed Safety Capsule • Eight 8.0 m Rotors • 6.0 MW • T/W ≥ 1.5

Main body 5 m • Capsule rise ≈ 1 m • Overall 24 × 24 × 6 m

Revision 00-T14-R8-6MW-B • Document-quality target: 9.5/10

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PROPRIETARY CONCEPT INFORMATION — Concept / Pre-Feasibility — Not Certified

Controlled baseline (locked): Design mass 14,000 kg with full fuel; 8 × 8.0 m rotors distributed around central capsule (no rotor above occupied volume); installed power 6.0 MW (6 × 1000 kW-class); real hover lift capability target 21–22 t → T/W ≥ 1.5; fuel 2,000 L; envelope 24 × 24 × 6 m; main body height 5 m; capsule rise ≈ 1 m; fixed non-jettisonable central capsule; 7 external mechanical damper legs; 6 capsule-side isolation dampers; ten-layer safety architecture.

This revision tightens mass allocation, makes power/FM assumptions explicit, records supersession of the prior 7×6.0 m baseline, and separates controlled values from calculated estimates and open items. Document-quality target 9.5/10 refers to configuration discipline and evidence control — not to aircraft feasibility or certification.

1. Document Control

Item	Value
Document Number / Title	00 — 14-Ton Helicopter 8×8 Controlled Baseline
Revision	00-T14-R8-6MW-B
Supersedes	00-T14-R8-6MW (prior draft) and 00-T10-R7-9.7 (7×6.0 m)
Configuration ID	GEN3-SH-14T-R8-6MW
Design mass (full fuel)	14,000 kg [T]
Rotors	8 × 8.0 m, distributed, no central-roof rotor [T]
Installed power	6.0 MW (6 × 1000 kW-class) [T]
Target real lift / T/W	21–22 t / ≥ 1.5 at 14 t [T]/[C-A]
Envelope	24 m × 24 m × 6 m [T]
Main body / capsule rise	5 m / ≈ 1 m [T]
Landing / capsule dampers	7 external + 6 capsule-side [T]
Capsule	Fixed, non-jettisonable, central [T]
Founder	Amirbahador Atai
Maturity	CONCEPT / PRE-FEASIBILITY
Certification	Not certified; compliance not established

Status codes: [T] design target/controlled; [C] calculated from supported inputs; [C-A] calculated with stated assumptions; [AV] awaiting validation; [OPEN] unresolved.

2. Executive Summary

This baseline is sized so that at a full-fuel design mass of 14,000 kg the aircraft has a real hover lift capability of approximately 21–22 tons, giving a thrust-to-weight ratio of at least 1.5. That objective drove the move from the earlier 7 × 6.0 m / 2.5–2.8 MW configuration ($T/W \approx 1.05\text{--}1.15$) to eight 8.0 m rotors and 6.0 MW installed power.

- Mass: 14,000 kg full fuel [T]. Mass status: PARTIAL (planning allocation closed toward 14 t; component data still [AV]).
- Rotors: 8 × 8.0 m, total disk area $\approx 402.1\text{ m}^2$, disk loading $\approx 34.8\text{ kg/m}^2$ at 14 t.
- Power: 6.0 MW installed (6 × 1000 kW-class). Approx. real power for 21–22 t lift $\approx 4.65\text{--}5.0\text{ MW}$ at $FM \approx 0.72$ [C-A]; installed power provides ~20–30% system margin.
- Geometry: 24 × 24 × 6 m; main body 5 m; capsule rise $\approx 1\text{ m}$.
- Safety: fixed central capsule, 7+6 damper system, ten-layer architecture, 8-parachute recovery concept — all retained from prior safety baseline.
- Explicit non-claims: not certified; performance not demonstrated; mass not closed by weigh; FM and installation losses are assumptions until tested.

3. Supersession and Comparison with Prior Baseline

Parameter	Prior 7×6.0 m (R7-9.7)	This baseline (R8-6MW-B)	Rationale
Rotors	7 × 6.0 m	8 × 8.0 m	Increase disk area for T/W
Disk area	≈ 198 m²	≈ 402 m²	+103%
Installed power	2.5–2.8 MW	6.0 MW	Enable 21–22 t real lift
Design mass	10 t class	14,000 kg full fuel	Realistic with larger system
Approx. T/W	1.05–1.15	≥ 1.5 (target)	Design driver
Fuel	1,700 L	2,000 L	Endurance
Envelope	17×17×5 m	24×24×6 m	Rotor clearance
Body / capsule rise	4 m / ~1 m	5 m / ≈ 1 m	Packaging
Landing legs	7	7	Unchanged
Capsule dampers	6	6	Unchanged
Capsule type	Fixed central	Fixed central	Unchanged
Ten-layer safety	Yes	Yes	Unchanged

Prior 7×6.0 m baseline is superseded for this product line. Safety architecture and capsule/landing philosophy are intentionally continuous.

4. Controlled Baseline (Locked)

Parameter	Value	Status
Design mass (full fuel)	14,000 kg	[T]
Rotor count / diameter	8 / 8.0 m (equal)	[T]
Rotor arrangement	Around capsule; no roof rotor above cabin	[T]
Total disk area	≈ 402.1 m²	[C]
Disk loading @ 14 t	≈ 34.8 kg/m²	[C]
Installed power	6.0 MW	[T]
Prime movers	6 × 1000 kW-class turboshaft	[T]
FM assumption (hover)	0.72	[A] until tested
Approx. real power @ 14 t	≈ 2.53 MW	[C-A]
Approx. real power for 21 t lift	≈ 4.65 MW	[C-A]
Approx. real power for 22 t lift	≈ 5.00 MW	[C-A]
Target real lift capability	21–22 tons	[T]/[C-A]
Target T/W @ 14 t	≥ 1.5	[T]
Fuel capacity / mass	2,000 L / ≈ 1,600 kg	[T] / [C]
Overall envelope	24 × 24 × 6 m	[T]
Main body height	5 m	[T]
Capsule rise	≈ 1 m	[T]
External legs / capsule dampers	7 / 6	[T]
Capsule	Fixed, non-jettisonable, central	[T]
Occupants	7	[T]
Safety architecture	Ten-layer concept	[T]
Recovery	8-parachute aircraft-level concept	[T]

5. Explicit Assumptions and Calculation Basis

Hover power estimates use actuator-disk / momentum theory with an assumed figure of merit $FM \approx 0.72$ at sea-level ISA density (1.225 kg/m^3). Installation losses, electrical conversion, transmission, download, and control reserves are not fully modelled; the 6.0 MW installed target is intended to cover approximately 20–30% margin above the 4.65–5.0 MW real-power band for 21–22 t lift. These are engineering assumptions [A] / [C-A], not verified performance.

Mass allocation is a planning budget closed toward 14,000 kg full fuel. Component masses remain [AV] until supplier data and weighed hardware exist. Envelope 24×24×6 m assumes comfortable tip clearance for 8.0 m rotors; exact station coordinates remain [AV] pending packaging study.

Assumption ID	Statement	Impact if wrong
A-FM-01	$FM \approx 0.72$ in hover	Real power required rises; T/W margin shrinks
A-INST-01	6.0 MW yields ~20–30% margin over 4.65–5.0 MW	May need more power or less mass
A-MASS-01	Full-fuel mass closable at 14,000 kg	T/W falls if mass grows
A-ENV-01	24×24×6 m provides adequate rotor clearance	Envelope may grow
A-MAT-01	High CFRP content supports mass target	Mass may grow if fraction not achieved

6. Vehicle Architecture

Central fixed safety capsule; main body height 5 m; capsule rises ≈ 1 m (overall height 6 m). Eight 8.0 m rotors distributed around the capsule with no primary lift rotor on the central roof above the occupied volume. Six 1000 kW-class turboshaft units feed a segmented distribution system to eight independent rotor channels. Seven external mechanical damper landing legs and six capsule-side isolation dampers form the primary energy-absorption chain. Fuel and hot machinery are separated from the occupant volume by containment zones.

7. Ten-Layer Safety Architecture (Retained)

#	Layer	Concept
1	Occupant survival capsule	Central fixed capsule, multi-angle CFRP roll-cage
2	Multi-stage energy absorption	Legs → airframe → capsule mounts → roll-cage → soft seats
3	Capsule–airframe isolation	High-stroke mounts + 6 capsule-side dampers
4	Distributed propulsion	8 independent rotor channels
5	Prime-mover redundancy	6 independent 1000 kW-class units
6	Fire detection/isolation/suppression	Zones, isolation, suppression at capsule perimeter
7	Control redundancy	Segregated sense/compute/actuate paths
8	Emergency descent/recovery	8-parachute concept + degraded modes
9	Water survival	Flotation concept, multi-orientation egress
10	Containment & modular isolation	Hazards outside cabin; isolate failed/burning rotor

No layer is a guaranteed survival device. Claims remain conceptual until analysis and test evidence exist.

8. Fixed Safety Capsule

Feature	Specification
Position	Central; rises ≈ 1 m above 5 m main body
Height contribution	Body 5 m + rise ≈ 1 m → overall 6 m
Structure	Near-spherical volume; multi-angle CFRP roll-cage
Seats	7 soft damper-mounted seats
Isolation	High-stroke mounts to airframe + 6 capsule-side dampers
Landing path	7 external mechanical damper legs
Fire	Perimeter suppression nozzles
Egress	Multiple exits; objective: usable after power loss
Recovery	Part of 8-parachute aircraft-level concept
Flight independence	None — fixed non-jettisonable

9. Rotor and Power Summary

Item	Value	Status
Rotors	8 × 8.0 m	[T]
Disk area total	≈ 402.1 m²	[C]
Disk loading @ 14 t	≈ 34.8 kg/m²	[C]
Thrust required @ 14 t	≈ 137.3 kN total	[C]
Ideal induced power @ 14 t	≈ 1.82 MW	[C]
Real power @ 14 t (FM 0.72)	≈ 2.53 MW	[C-A]
Real power for 21 t lift	≈ 4.65 MW	[C-A]
Real power for 22 t lift	≈ 5.00 MW	[C-A]
Installed power	6.0 MW	[T]
Prime movers	6 × 1000 kW-class	[T]
Target real lift	21–22 t	[T]/[C-A]
Target T/W @ 14 t	≥ 1.5	[T]

Power flow concept: turboshaft → generator → segmented HV bus → independent inverters → rotor motors. Channel independence is a safety requirement, not yet demonstrated.

10. Mass Budget (Closed Toward 14 t)

Group	Allocation (kg)	Notes
Primary airframe & rotor support	2,750	Planning; high-CFRP target
Capsule, roll-cage, isolators, seats	1,250	Planning
8 rotors, hubs, controls (8.0 m)	1,550	Supplier-dependent [AV]
Powerplant, generators, drives (6 MW)	2,600	Six 1000 kW-class [AV]
Fuel system (dry)	330	Planning
Fuel 2,000 L	1,600	0.80 kg/L [C]
Avionics, flight control, electrical	700	Redundancy included
Landing gear 7+6 dampers	550	Planning
Fire, recovery, safety equipment	650	8-parachute concept
Thermal management	550	6 MW class
Interconnects, fluids, growth	1,220	Margin to 14,000
Total (full fuel)	14,000	Controlled target [T]

Mass status: PARTIAL. Sum forced to 14,000 kg as controlled target. Several lines remain [AV] until supplier and process data exist. Growth above 14 t directly erodes T/W.

11. Fuel, Endurance and Range (Order-of-Magnitude)

Parameter	Value	Status
Fuel capacity	2,000 L	[T]
Fuel mass	≈ 1,600 kg	[C]
Hover endurance @ 14 t	~90–120 min	Order-of-magnitude [C-A]
Cruise endurance	~3.0–4.0 h	Order-of-magnitude [C-A]
Range @ ~200 km/h	~600–800 km	Order-of-magnitude [C-A]

Not certified performance. Actual values depend on SFC, reserves, mission and atmosphere.

12. Flight Control, Structure, Emergency

Eight independent rotor channels and six prime movers support graceful degradation. Structure uses multiple load paths around the capsule; 7+6 damper elements are the primary impact isolation path. Emergency matrix: single rotor → reconfiguration; multiple degradation → emergency mode + 8-parachute concept; fire → isolate and recover; water → capsule survival/egress concept. No guarantee of intact airframe in all water impacts.

13. Verification, Certification and Risk

Verification methods: analysis, inspection, test, demonstration. Planned ≠ completed. Simulation ≠ test. This document does not claim certification. Required work includes FHA/PSSA/SSA, common-cause analysis, structural and damper tests, rotor performance tests, and thermal validation of the 6 MW package.

Risk	Impact	Mitigation direction
Mass > 14 t	High — T/W drops	Supplier mass + CFRP process control
Real lift < 21 t	High — T/W < 1.5	Rotor CFD, FM test, power margin
FM < 0.72	High	Blade design + test
Rotor/hub mass growth	High	Early supplier engagement
Thermal shortfall at 6 MW	High	Cooling design + test
Control of 8 large rotors	Medium-High	HIL + progressive envelope
Damper energy shortfall	High	Full-scale impact articles
Envelope clearance inadequate	Medium	Packaging study → freeze 24×24×6

14. Open Items Before Rev 01

- Mission profile freeze (payload, speed, altitude, range, reserves).
- 8.0 m rotor aero model (blades, solidity, tip speed, FM validation plan).
- OEM selection and generator matching for 6 × 1000 kW-class.
- Component-level mass statement; close mass at 14,000 kg.
- Capsule load cases and damper stroke/force targets.
- Parachute sizing and attachment loads.
- Water-impact and flotation requirements.
- Formal safety architecture and failure database.
- Certification basis mapping.
- Rotor station coordinates and clearance confirmation for 24×24×6 m.

15. Design Rules

1. 14,000 kg full-fuel is a controlled target — not optional paper mass.
2. $T/W \geq 1.5$ at 14 t is a controlled design driver.
3. Do not claim safety or performance credit without evidence.
4. Prefer passive mechanical safety layers where practical.
5. No silent baseline change; document vs configuration revision kept separate.
6. Assumptions (FM, installation margin, CFRP fraction) stay labelled until verified.
7. Main body 5 m and capsule rise ≈ 1 m remain controlled geometry.

16. Performance Envelope (Preliminary)

Metric	Value	Status
Design mass full fuel	14,000 kg	[T]
Disk area	≈ 402.1 m²	[C]
Disk loading @ 14 t	≈ 34.8 kg/m²	[C]
Ideal power @ 14 t	≈ 1.82 MW	[C]
Real power @ 14 t (FM 0.72)	≈ 2.53 MW	[C-A]
Real power for 21 t / 22 t lift	≈ 4.65 / 5.00 MW	[C-A]
Installed power	6.0 MW	[T]
Target real lift	21–22 t	[T]/[C-A]
Target T/W @ 14 t	≥ 1.5	[T]
Fuel	2,000 L / ≈ 1,600 kg	[T]/[C]
Envelope	24 × 24 × 6 m	[T]
Body height / capsule rise	5 m / ≈ 1 m	[T]
Hover endurance (oom)	~90–120 min	[C-A]
Cruise endurance (oom)	~3–4 h	[C-A]
Range oom @ 200 km/h	~600–800 km	[C-A]

oom = order-of-magnitude. Not certified. Not demonstrated.

17. Final Baseline Statement

Document 00 Revision **00-T14-R8-6MW-B** (14-Ton Helicopter 8×8) locks the following controlled baseline: full-fuel design mass **14,000 kg**; **eight 8.0 m** rotors distributed around a fixed central safety capsule with no primary lift rotor above the occupied volume; installed power **6.0 MW** (six 1000 kW-class turboshaft units); target real hover lift **21–22 tons** and thrust-to-weight ≥ 1.5 at 14 t; fuel **2,000 L**; envelope **24 × 24 × 6 m**; main body height **5 m**; capsule rise $\approx 1\text{ m}$; seven external mechanical damper legs; six capsule-side isolation dampers; ten-layer safety architecture; eight-parachute aircraft-level recovery concept.

Power and lift figures that depend on $FM \approx 0.72$ and installation margin are labelled [C-A]. Mass is allocated to sum to 14,000 kg but remains PARTIAL until component evidence exists. The prior 7×6.0 m baseline is superseded for T/W reasons; safety capsule and layered protection philosophy are continuous. This document does not claim the aircraft is feasible, safe, certifiable or production-ready. Engineering truth has priority over document completeness.

Founder / Principal Author: **Amirbahador Atai**

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Status: Concept / Pre-Feasibility — Not Certified

END OF DOCUMENT 00 — 14 t / 8×8.0 m / 6 MW / T/W ≥ 1.5